Dhruv Ranjan

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Summary

Computer Engineering undergraduate who ships production backend systems, including a mock-test platform live on AWS. Completed **NVIDIA Conversational AI** coursework (**NLP**, **generative AI**); solved **700+ DSA problems**.

Technical Skills

Languages: C/C++, Python, JavaScript, TypeScript, SQL
Web & Backend: Node.js, React.js, FastAPI, Prisma, Mongoose, REST APIs, HTML, CSS
Databases: PostgreSQL, MySQL, MongoDB, Redis
Cloud & DevOps: AWS (EC2, ECR, SSM), Docker, Kubernetes, GitHub Actions (CI/CD), Linux, Git, Postman
AI / LLM: RAG, LLM APIs (Claude), embeddings, BM25 + vector hybrid search, ChromaDB
Core CS: DSA, OOP, DBMS, Operating Systems, Computer Networks, System Design

Experience

Software Developer (Freelance) | *Entrance UG ↗* | *React, Node.js, PostgreSQL, AWS* June 2026 – Aug. 2026

- Built an IPMAT/CUET exam-prep platform used by **500+ students**: timed mocks, lectures, daily RC drills and mentorship, with role-based portals (student, parent, mentor, admin) on one Postgres schema (100+ tables)
- Built the payments layer so money and access can never disagree — gateway-verified capture, idempotent settlement, hourly reconciliation — proven across **29 payment scenarios**
- Built and instrumented the API's **CI/CD pipeline on AWS** — GitHub Actions → ECR → SSM onto EC2 — then **cut image build 40%** (1m50s to 1m06s) by caching the dependency layer

Full Stack Intern | *Enerbuff Solutions* | *Node.js, Postman* Aug. 2026 – Present

- Mapped the Beckn/ONIX BAP–BPP request–response flow for the company's India Energy Stack (IES) integration and verified each step against the IES sandbox in Postman

Project Intern | *IIT Kanpur* May 2025 – Jan 2026

- Designed the normalized relational schema for a production e-commerce platform, with ACID guarantees
- Cut bulk catalog API latency to <**50ms** by rewriting query plans, adding Redis caching and fixing index coverage
- Made orders concurrency-safe via **row-level locking** and **idempotency keys**, stopping duplicates and overselling

Projects

ZoneWise: Location-Aware RAG | *Python, FastAPI, React, ChromaDB, Groq* | Live | GitHub Oct. 2026

- Built and deployed a **geospatial RAG assistant** for Delhi's Master Plan 2047 that maps a clicked location to official zones and answers with clause-level citations
- Raised top-5 retrieval recall from **55% to 71%** by adding zone-aware filtering to hybrid BM25 + vector search, doubling it on location-specific queries (**32%** → **64%**)
- Grounded LLM answers with schema-validated citations and a **claim-verification** step; shipped on Azure via Docker and GitHub Actions CI (640 tests)

Chess Engine | *Python* | GitHub Feb. 2026 – Mar. 2026

- **Cut search time 93%** over 13 benchmark positions and gained 2 plies of depth at one second per move, by adding move ordering, an incrementally hashed transposition table, quiescence search with static exchange evaluation, and principal variation search

Certifications

NVIDIA DLI: Accelerating End-to-End Data Science Workflows ↗ (Dec. 2025): GPU-accelerated ML with cuDF, cuML

Education

Thapar Institute of Engineering and Technology Aug. 2023 – May 2027
Bachelor of Technology in Computer Engineering CGPA: 8.3/10 *Patiala, Punjab*

- Elective: NVIDIA Conversational AI (data science, NLP, speech processing, generative AI)

Methodist High School | *ISC (Class XII): 89.4% (2023)* | *ICSE (Class X): 93.6% (2021)* *Kanpur, Uttar Pradesh*